EXPERIMENT 5

Wireshark Packet Capture and Analysis

1. Aim

To capture and examine network packets with the Wireshark packet analysis tool, use protocol filters for TCP, UDP, ARP, DNS, HTTP, IP, ICMP, and DHCP, and study the captured data.

2. Description

Wireshark is a tool for capturing and analysing network traffic. It records packets in real time and presents their contents in a readable format. It helps users examine network traffic, review protocol information, filter packets, troubleshoot network issues, and understand how devices communicate.

The tool records packets transmitted and received by a computer. These packets can be examined in the Packet List, Packet Details, and Packet Bytes panels.

3. Packet Capture

A suitable network interface was chosen in Wireshark, and the capture was set to stop automatically after 100 packets. The captured data was saved for later examination.

The capture was filtered to show packets associated with the following protocols:

- TCP/UDP
- ARP
- DNS
- HTTP
- IP/ICMP
- DHCP

Flow graphs were also created to observe communication involving TCP/UDP and DNS packets.

4. Observation

Wireshark was used to capture packets and examine communication between network devices. In promiscuous mode, a network interface may capture packets travelling across the network even when those packets are not specifically addressed to the computer.

ARP packets do not include a transport-layer header, since ARP does not use transport protocols such as TCP or UDP. Its purpose is to find the MAC address linked to a given IP address.

DNS usually uses UDP at the transport layer and commonly communicates through port 53, although TCP can be used in some cases.

HTTP uses port 80 as its default port number.

A broadcast IP address sends packets to all devices on a specific network. For instance, for the 192.168.1.0/24 network, the broadcast address is 192.168.1.255. The observation topics include promiscuous mode, ARP headers, the DNS transport protocol, the HTTP port, and broadcast addressing.

5. Outcome

Network packets were captured and examined successfully with Wireshark. Packets from different protocols were filtered and reviewed, and the communication flow between devices was studied using the packet analysis features.

6. Challenges Faced

- Choosing the appropriate network interface for capturing packets.
- Finding packets for the required protocols within a large capture.
- Using the correct filters for TCP, UDP, ARP, DNS, HTTP, ICMP/IP, and DHCP traffic.

• Understanding the various fields shown in packet details.

• Creating and interpreting the flow graph.

• Capturing the required number of packets without errors.

7. Result

Therefore, Wireshark was used successfully to capture and analyse network packets. Packets from different protocols were filtered, inspected, and studied.